Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	404notFound
Project / Solution Name	StockSenseAI
Name (Leader)	Haemaavadhaana Seemadaran
Email & Phone (Leader)	Email: haemaavadhaana@gmail.com Phone: 0182854952
University Affiliation(s)	Asia Pacific University of Technology & Innovation
Live Prototype URL	https://stock-sense-ai-Haemaa.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/haemaavadhaana123/StockSenseAI.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

StockSense AI is a Python-based inventory management and replenishment application built using Streamlit. Users interact with the application through a web-based interface to upload inventory data in CSV format, view stock levels, identify low-stock products, and receive replenishment recommendations.

The application processes inventory data using reusable Python modules and pandas for CSV data handling and data processing. Inventory validation and stock classification logic determine product stock conditions and calculate suggested replenishment quantities. Users can create replenishment requests, review and approve or reject requests, and track workflow activities through the Approval Centre and Activity Log.

Inventory and workflow data are stored in CSV files. The prototype runs in a Replit development environment and uses local file-based storage rather than a separate database. The current implementation uses rule-based inventory and replenishment logic; an external AI model API is not required for its core functionality. Orbyte integration has not yet been demonstrated.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python — version used in the project	Main programming language for application logic and workflow processing	Python Software Foundation License
Streamlit — version used in the project	Web interface, dashboard, inventory management and workflow pages	Apache License 2.0
pandas — version used in the project	CSV loading, tabular data processing and inventory calculations	BSD 3-Clause License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
No external API is currently called by the core inventory and replenishment functionality.	N/A	No external API is currently called by the core inventory and replenishment functionality.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment and running the Streamlit prototype.	Actual Replit plan
GitHub	Source-code hosting and version control for the project repository.	Actual account plan

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None — no AI model integrated into the running application	N/A	The application uses rule-based inventory and replenishment logic rather than a trained AI model.	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
StockSense AI Sample Inventory	Created for the project; local CSV file	Synthetic	Project-created; no separate dataset license	No
Custom SME Inventory Test Dataset (test_sme_inventory.csv)	Created for project testing	Synthetic	Project-created; no separate dataset license	No
Replenishment Requests and Activity Log	Generated by the application during workflow testing	Synthetic / self-generated	Project-created; no separate dataset license	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT (GPT-6)	Assistance with Python code development, debugging, application logic, testing guidance, and technical documentation	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted code and documentation were reviewed by the team and tested through hands-on execution of the StockSense AI prototype. Testing covered custom CSV inventory uploads, stock classification, dashboard updates, replenishment recommendations, request creation, approval or rejection, and activity logging. Errors encountered during development were investigated and corrected. The team evaluated the application's behaviour against expected results and remains responsible for the accuracy, functionality, security, and final submission of all project materials.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ◉ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No official SDC sponsor-provided resources were used in the development of StockSense AI. The prototype was developed using the team's own development environment and tools. Its core inventory management, stock classification, replenishment recommendation, and approval workflows rely on locally implemented application logic and CSV-based data storage. The application does not depend on sponsor-provided APIs, cloud credits, or datasets for its core functionality.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

StockSense AI processes inventory data, including product IDs, product names, categories, stock quantities, units, and supplier names. The prototype also stores replenishment requests and activity logs generated during workflow operations. The test inventory uses synthetic data and does not contain personal or sensitive information. The application does not intentionally collect personal user data or send inventory data to external AI APIs. Inventory and workflow records are stored in CSV files within the application's working environment. Custom inventory uploads are handled using temporary session-based files. Data retention depends on the storage location and session lifecycle. Users should avoid uploading confidential business information or personal data until appropriate access controls, secure persistent storage, and a formal data-retention policy are implemented.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

StockSense AI currently uses rule-based inventory classification and replenishment calculations rather than a trained AI model. Recommendations depend on the accuracy and completeness of the uploaded inventory data and may not account for demand forecasts, supplier lead times, seasonal changes, or market conditions. Invalid or incomplete CSV data may prevent processing. The prototype uses CSV-based file storage rather than a production database, which limits scalability and multi-user data management. Custom inventory uploads use session-based temporary files, so uploaded data may not persist across sessions. Human approval is required before simulated procurement actions, and the prototype does not place real purchase orders. These limitations are mitigated through inventory validation, automated stock classification, workflow logging, and human review before procurement decisions.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

23rd October 2026